JPE Replication Report

PUBLISHED
July 28, 2026

1 Paper Identification

ID	20250186
Author	Kate Smith
Title	The Welfare Effects of Price Shocks and Household Relief Packages: Evidence from an Energy Crisis
Iteration	1

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

1. You are required to cite all raw data in both paper and readme.
2. All exhibits are reproduced correctly, which is a success! Congratulations.

3 General

- [x] Data Availability and Provenance Statements
  - [x] Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - [ ] Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - [x] License for Data (optional, but recommended)
  - [x] Details on each Data Source
- [x] Dataset list
- [x] Computational requirements
  - [x] Software Requirements
  - [x] Controlled Randomness (as necessary)
  - [x] Memory, Runtime, and Storage Requirements
- [x] Description of programs/code
  - [x] License for Code (Optional, but recommended)
- [x] Instructions to Replicators
  - [x] Details (as necessary)
- [x] List of tables and programs
- [x] References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- [x] uses observational data.
- [x] uses *confidential* observational data.
- [ ] uses data generated via experiment or trial by the authors (in field or lab).
  - [ ] For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.

- [ ] A PDF copy of the instructions given to participants in both original language and an English translation is included.
- [ ] uses data generated via survey by the authors (in field or online)
  - [ ] The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- [ ] is a simulation study: the data are generated by an algorithm.
- [ ] is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

> {REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

> {REQUIRED} Please provide affirmation of support for replication checks. - As per the JPE policy, "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

> {REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
ExactOne bank transaction data	no	yes	-	yes	yes	no
Living Costs and Food Survey (LCFS)	no	yes	-	yes	yes	no
ONS CPI microdata	yes	no	yes	yes	yes	no
ONS CPI	yes	no	-	yes	yes	no
ONS population data	yes	no	yes	yes	yes	no
Ofgem price cap data	yes	no	yes	yes	yes	no
BEIS energy unit cost and expenditure data	yes	no	yes	yes	yes	no
Department for Energy Security and Net Zero data	yes	no	yes	yes	yes	no
Met Office weather data	yes	no	-	yes	yes	no
Geographic lookup files	yes	no	yes	yes	yes	no
ONS Consumer trends data	yes	no	yes	yes	yes	no

6 Requirements

- [x] README is in TXT, MD, or PDF format
- [x] README is accessible at the root of the package (not inside a folder)
- [ ] Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)

- [ ] Authors (with affiliations) are listed in the same order as on the paper

> {REQUIRED} Please review the title of the deposit as per our guidelines.

> {REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

6.1 Stated Requirements

- [ ] No requirements specified
- [x] Software Requirements specified as follows:
  - Stata version 18 or later (with packages: binscatter, cdfplot, cleanplots, coefplot, confirmdir, distinct, estout, labutil, listtab, numdate)
  - MATLAB R2024b (with Optimization Toolbox)
  - R version 4.4.3 (with packages: tidyverse, sf, readxl)
- [x] Computational Requirements specified as follows:
  - 5 GB working storage
- [x] Time Requirements specified as follows:
  - Under 1 hour for Stages 2–3. Under 1 hour for MATLAB estimation and simulations. Synthetic data generator runs in 5–10 minutes.
- [x] Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 33
- Variables flagged in data: 33
- Code files with PII references: 62
- PII references in code: 1005

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`availabletariffs_Ofgem.xlsx`	1	son
Data	`humidity2018.csv`	1	name
Data	`humidity2019.csv`	1	name
Data	`humidity2020.csv`	1	name
Data	`humidity2021.csv`	1	name
Data	`humidity2022.csv`	1	name
Data	`humidity2023.csv`	1	name
Data	`postcodedistricts2ofgemregions_expanded.csv`	1	district
Data	`rainfall2018.csv`	1	name
Data	`rainfall2019.csv`	1	name
Data	`rainfall2020.csv`	1	name
Data	`rainfall2021.csv`	1	name

File Type	File	Variables/References	PII Categories
Data	`rainfall2022.csv`	1	name
Data	`rainfall2023.csv`	1	name
Data	`tempav2018.csv`	1	name
Data	`tempav2019.csv`	1	name
Data	`tempav2020.csv`	1	name
Data	`tempav2021.csv`	1	name
Data	`tempav2022.csv`	1	name
Data	`tempav2023.csv`	1	name

See Appendix for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-07-17T16:12:29.906

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
95	1	31	1	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/CompVar.m	0.0	209f5132f40eacbf9f742413d069577a9a2a4f37
/ReplicationPackage/Programs/5.modelSimulations/Simulations/parammap.m	0.0	f87abf94c2b57aa008f8fba09f3bb941b43b6364
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/parammap.m	0.0	f87abf94c2b57aa008f8fba09f3bb941b43b6364
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/Classical.m	0.0	dc93cdd167a77ef2f2d7d4b5f088a6d9f83022a9
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/drawcoefficients.m	0.0	94e1838a76ac49ebd28cff224591cce48ee5ae23
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/rebatesort.m	0.0	90b0d44cc533aa365822876da24de12501b7ebf2
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/Hicksian.m	0.0	bdb34dbd59311459a0be4e0dd5ac7268cbfde183
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/Marshbehave.m	0.0	69dff503efaab35fbc978e179d35fc880c44870c
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/dataprepare.m	0.0	29666cc0497d7c640824147a5f27811317870e6a
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/equivtransfer.m	0.0	4dc18e5168e529aa74cd4086211e517878ae81c6
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/equivpubliccost.m	0.0	da0c69effa9b15083232fd9e832cecf051f551a7
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/equivrebate.m	0.0	14a06ea043d42020add55b9701619d4f2fcb09be
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/Behavioural.m	0.0	83c9148b8c5ef61aa9514e1930558e696c415953
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/MPCE.m	0.0	0fc0057c53a10a350840dedb17f1c50ba8b2a3c6

Filename	Size (MB)	Checksum (MD5)
/ReplicationPackage/Programs/5.modelSimulations/SimulationsSynthetic/Observedpolicy.m	0.0	8dd6c6c79f6ee191e602c723d7642c7290fb3edc

7.1.4 Zero Size Files Report

We found the following zero size files:

Filename	Size (MB)	Checksum (MD5)
/ReplicationPackage/Programs/1.setupWeatherR/.Rproj.user/shared/notebooks/patch-chunk-names	0.0	da39a3ee5e6b4b0d3255bfef95601890afd80709

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/ReplicationPackage/dataInput/dataAnalysis/observedpolicy.raw	131.97
/ReplicationPackage/dataInput/dataAnalysis/restrictions.raw	163.66
/ReplicationPackage/dataInput/dataAnalysis/counterfactualpolicy_incadj.raw	1544.74
/ReplicationPackage/dataInput/dataAnalysis/counterfactualpolicy.raw	2057.46
/ReplicationPackage/dataInput/dataAnalysis/sensitivity1.raw	2142.92
/ReplicationPackage/dataInput/dataAnalysis/sensitivity2.raw	2142.92

7.2 Code description

There are 35 Stata `.do` files and 58 Matlab `.m` files provided in the replication package, including a main `master.do` file.

cloc		github.com/AlDanial/cloc v 2.06 T=1.37 s (81.8 files/s, 946198.7 lines/s)		
Language	**files**	**blank**	**comment**	**code**
CSV	31	0	0	1281169
Stata	35	3122	67	8201
MATLAB	43	593	254	1610
Markdown	1	96	0	363
R	2	18	16	47
——–	——–	——–	——–	——–
SUM:	112	3829	337	1291390

- [x] The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

> NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos 16v CPU, 64 GB RAM
- StataNow 19.5
- Matlab R2024b

- R 4.3.1

8 Replication

- Downloaded code and data.
- Read README.md and followed instructions for Option A — Using synthetic data.
- Set globals in Programs/master.do, Programs/4.modelEstimates/Estimation/Run.m, and Programs/5.modelSimulations/SimulationsSynthetic/Run.m.
- Ran Programs/master.do in Stata.
- Code ran to completion.
- Commited to git repo.

9 Findings

9.1 Data Preparation Code

- Program `Programs/master.do` referring to all other programs ran without any errors.

9.2 Tables and Figures

Data Preparation Program	Dataset created	Reproduced?
`synthetic_data_generator.do`	`$dataProcess/CSenergyallspending_step5.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/CSenergyallspending_step6.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/CSenergyallspending_step2b_weights.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/CSenergy_step1_combine.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/heterogeneity_measures.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/preestimationdata.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/lcfsdata.dta`	Yes
`synthetic_data_generator.do`	`$dataProcess/lcfsdata_incAHC.dta`	Yes
`2.setupData/0.cpi.do`	`$dataProcess/cpi_stata_annual`	Yes
`2.setupData/0.cpi.do`	`$dataProcess/cpi_stata`	Yes
`2.setupData/1.importenergypricecap.do`	`$dataProcess/beis_averageunitcosts_elec`	Yes
`2.setupData/1.importenergypricecap.do`	`$dataProcess/beis_averageunitcosts_gas`	Yes
`2.setupData/1.importenergypricecap.do`	`$dataProcess/OfgemPriceCaps`	Yes
`2.setupData/1.importenergypricecap.do`	`$dataProcess/OfgemPriceCaps_EPG`	Yes
`2.setupData/1.importenergypricecap.do`	`$dataProcess/OfgemPriceCaps_EPG_monthly`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/equiv`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/lcfsdemogs`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/lcfscpiexpends`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/lcfsrpiexpends`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/lcfsincome`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/lcfsdata`	Yes
`2.setupData/3.LCFS.do`	`$dataProcess/LCFS_sharegas_byincandexp`	Yes
`2.setupData/6.insheetweatherprices.do`	`$dataProcess/laspeyres_hetweights`	Yes

Data Preparation Program	Dataset created	Reproduced?
`2.setupData/8.analysis.do`	`$dataProcess/CSenergyallspending_step5`	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
Figure 2.1: Energy price cap, energy price guarantee and cheapest available pricing plans	`3.descriptiveResults/3.describepricecap.do`	`FIG_energypricecap_epg_cheapest.pdf` (`3.descriptiveResults/3.describepricecap.do:255`)	Yes
Figure 3.1: Energy spending across the income distribution	`3.descriptiveResults/4.exposure.do`	`FIG_x_ener_Vs_inc_2019_20.pdf` (`3.descriptiveResults/4.exposure.do:84`), `FIG_shr_ener_Vs_inc_2019_20.pdf` (`3.descriptiveResults/4.exposure.do:102`)	Yes
Figure 4.1: Log energy spending over the crisis	`3.descriptiveResults/5.priceresponse.do`	`FIG_lx_ds.pdf` (`3.descriptiveResults/5.priceresponse.do:172`)	Yes
Table 4.1: Energy price elasticities	`3.descriptiveResults/5.priceresponse.do`	`TAB_elas_apr22.tex` (`3.descriptiveResults/5.priceresponse.do:302, 304`), `TAB_elas_apr22_se.tex` (`3.descriptiveResults/5.priceresponse.do:303, 305`), `TAB_qchg_apr22.tex` (`3.descriptiveResults/5.priceresponse.do:306, 308`), `TAB_qchg_apr22_se.tex` (`3.descriptiveResults/5.priceresponse.do:307, 309`)	Yes
Figure 4.2: Heterogeneity in energy price elasticities, by pre-shock energy spending and income	`3.descriptiveResults/5.priceresponse.do`	`FIG_elas_apr22_hom.pdf` (`3.descriptiveResults/5.priceresponse.do:881`)	Yes
Figure 4.3: Energy spending over the transfer period, prepayment households	`3.descriptiveResults/6.MPCs.do`	`FIG_lx_ds_rebates_qtr_PP_realvsfitted.pdf` (`3.descriptiveResults/6.MPCs.do:371`)	Yes
Table 4.2: Marginal propensity to consume energy out of transfers	`3.descriptiveResults/6.MPCs.do`	`TAB_MPCrebates_PP.tex` (`3.descriptiveResults/6.MPCs.do:556, 557, 558, 559`), `TAB_MPCrebates_PP_N.tex` (`3.descriptiveResults/6.MPCs.do:560`)	Yes
Figure 5.2: Winter energy demand in and out of sample	`4.modelEstimates/2.modelfit.do`	`FIG_outinsample_exp.pdf` (`4.modelEstimates/2.modelfit.do:213`), `FIG_outinsample_inc.pdf` (`4.modelEstimates/2.modelfit.do:224`)	Yes
Figure 6.1: Distribution of household losses by income	`5.modelSimulations/4.tablesfigures.do`	`FIG_Loss_level.pdf` (`5.modelSimulations/4.tablesfigures.do:26`), `FIG_Loss_proportional.pdf` (`5.modelSimulations/4.tablesfigures.do:35`)	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
Table 6.1: Average losses	`5.modelSimulations/4.tablesfigures.do`	`losses.tex` (`5.modelSimulations/4.tablesfigures.do:69, 72`), `poverty.tex` (`5.modelSimulations/4.tablesfigures.do:75`)	Yes
Table 6.2: Efficiency costs	`5.modelSimulations/4.tablesfigures.do`	`efficiency.tex` (`5.modelSimulations/4.tablesfigures.do:84, 87`)	Yes
Figure 6.2: Counterfactual policy responses	`5.modelSimulations/4.tablesfigures.do`	`FIG_optimal_frontier_W.pdf` (`5.modelSimulations/4.tablesfigures.do:99`), `FIG_optimal_decomp.pdf` (`5.modelSimulations/4.tablesfigures.do:110`)	Yes
Appendix Figure A.1: Energy price cap, energy price guarantee and cheapest available price plans for prepay consumers	`3.descriptiveResults/3.describepricecap.do`	`FIG_energypricecap_epg_prepay.pdf` (`3.descriptiveResults/3.describepricecap.do:265`)	Yes
Appendix Figure A.2: Seasonality of energy spending by payment type	`3.descriptiveResults/1.summarystats.do`	`FIG_paymenttypes_season.pdf` (`3.descriptiveResults/1.summarystats.do:126`)	Yes
Appendix Figure A.3: Age and geographic sample composition	`3.descriptiveResults/1.summarystats.do`	`FIG_sample_gor.pdf` (`3.descriptiveResults/1.summarystats.do:179`), `FIG_sample_age.pdf` (`3.descriptiveResults/1.summarystats.do:161`)	Yes
Appendix Figure A.4: Non-durable and energy spending in ExactOne and Living Costs and Food Survey	`3.descriptiveResults/1.summarystats.do`	`FIG_lcfsvscs_nondur.pdf` (`3.descriptiveResults/1.summarystats.do:240`), `FIG_lcfsvscs_energy.pdf` (`3.descriptiveResults/1.summarystats.do:249`), `FIG_rankrank_energyvsnondur_LCFS.pdf` (`3.descriptiveResults/1.summarystats.do:272`), `FIG_rankrank_energyvsnondur_CS.pdf` (`3.descriptiveResults/1.summarystats.do:293`)	Yes
Appendix Figure A.5: Income and energy spend, by payment type	`3.descriptiveResults/1.summarystats.do`	`FIG_sample_inc_distr.pdf` (`3.descriptiveResults/1.summarystats.do:333`), `FIG_sample_eexp_distr.pdf` (`3.descriptiveResults/1.summarystats.do:349`)	Yes
Appendix Figure A.6: Trends in energy spending in ExactOne data and National Accounts	`3.descriptiveResults/2.macrotrends.do`	`FIG_NAvsCS_spend.pdf` (`3.descriptiveResults/2.macrotrends.do:196`)	Yes
Appendix Figure B.1: Energy spending across the income and total expenditure distributions	`3.descriptiveResults/4.exposure.do`	`FIG_x_ener_Vs_inc_2019.pdf` (`3.descriptiveResults/4.exposure.do:152`), `FIG_x_ener_Vs_texp.pdf` (`3.descriptiveResults/4.exposure.do:180`)	Yes
Table B.1: Explanatory power of household	`3.descriptiveResults/4.exposure.do`	`TAB_r2_LCFS_expcats.tex` (`3.descriptiveResults/4.exposure.do:270, 271`)	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
income and demographics, by spending categories			
Table B.2: Persistence of energy spending	`3.descriptiveResults/4.exposure.do`	`TAB_exp_AR.tex` (`3.descriptiveResults/4.exposure.do:393, 394, 395, 396`)	Yes
Appendix Table B.3: Persistence of energy spending by income quartile, 2019-20	`3.descriptiveResults/4.exposure.do`	`TAB_exp_AR_prebyinc.tex` (`3.descriptiveResults/4.exposure.do:399, 400, 401, 402`)	Yes
Appendix Table B.4: Persistence of energy spending by income quartile, 2019-23	`3.descriptiveResults/4.exposure.do`	`TAB_exp_AR_crisisbyinc.tex` (`3.descriptiveResults/4.exposure.do:399, 400, 401, 402`)	Yes
Appendix Figure C.1: Quantity share of total energy from electricity	`3.descriptiveResults/2.macrotrends.do`	`FIG_shr_elec.pdf` (`3.descriptiveResults/2.macrotrends.do:239`)	Yes
Appendix Figure C.2: Alternative price indexes	`3.descriptiveResults/3.describepricecap.do`	`FIG_altpriceindexes.pdf` (`3.descriptiveResults/2.macrotrends.do:251`)	Yes
Appendix Figure C.3: Share of energy spending on gas, by income quintile and energy spending quintile	`3.descriptiveResults/1.summarystats.do`	`FIG_sharegas_byincandexp.pdf` (`3.descriptiveResults/1.summarystats.do:794`)	Yes
Appendix Figure C.4: Energy prices and spending in Northern Ireland relative to the rest of the UK	`3.descriptiveResults/3.describepricecap.do`, `5.priceresponse.do`	`FIG_energypriceNI.pdf` (`3.descriptiveResults/3.describepricecap.do:125`), `FIG_lx_GBvsNI.pdf` (`3.descriptiveResults/5.priceresponse.do:166`)	Yes
Appendix Figure C.5: Heterogeneity in elasticities, estimated using pre-shock energy spending and income specific price index weights	`3.descriptiveResults/5.priceresponse.do`	`FIG_elas_apr22_het.pdf` (`3.descriptiveResults/5.priceresponse.do:881`)	Yes
Table C.1: Energy price elasticities, robustness	`3.descriptiveResults/5.priceresponse.do`	`TAB_qchg_apr22_app.tex` (`3.descriptiveResults/5.priceresponse.do:795`), `TAB_qchg_apr22_se_app.tex` (`3.descriptiveResults/5.priceresponse.do:796`), `TAB_elas_apr22_app.tex`	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
		(`3.descriptiveResults/5.priceresponse.do:793`), `TAB_elas_apr22_se_app.tex` (`3.descriptiveResults/5.priceresponse.do:794`), `TAB_N_apr22_app.tex` (`3.descriptiveResults/5.priceresponse.do:797`)	
Appendix Figure C.6: Energy spending over the transfer period, direct-debit households	`3.descriptiveResults/6.MPCs.do`	`FIG_lx_ds_rebates_qtr_DD_realvsfitted.pdf` (`3.descriptiveResults/6.MPCs.do:360`), `FIG_lx_ds_rebates_qtr_noPP.pdf` (`3.descriptiveResults/6.MPCs.do:350`)	Yes
Table C.2: MPCE estimates for transfers (households paying by direct debit)	`3.descriptiveResults/6.MPCs.do`	`TAB_MPCrebates_DD.tex` (`3.descriptiveResults/6.MPCs.do:562, 563, 564, 565`), `TAB_MPCrebates_DD_N.tex` (`3.descriptiveResults/6.MPCs.do:566`)	Yes
Table D.1: Parameter estimates	`4.modelEstimates/1.coefficients.do`	`coefficients.tex` (`4.modelEstimates/1.coefficients.do:82, 83, 84, 86, 87, 88`)	Yes
Appendix Figure D.1: Heterogeneity in elasticities, model-based estimates	`4.modelEstimates/2.modelfit.do`	`FIG_elashet_model.pdf` (`4.modelEstimates/2.modelfit.do:273`)	Yes
Appendix Figure D.2: Heterogeneity in Engel curves	`4.modelEstimates/2.modelfit.do`	`FIG_Engelcurve.pdf` (`4.modelEstimates/2.modelfit.do:297`)	Yes
Appendix Figure D.3: Winter energy demand jointly by income and pre-shock energy spending	`4.modelEstimates/2.modelfit.do`	`FIG_insample_inc.pdf` (`4.modelEstimates/2.modelfit.do:187`), `FIG_outofsample_inc.pdf` (`4.modelEstimates/2.modelfit.do:176`)	Yes
Appendix Figure D.4: Hold-out sample validation of flypaper effect	`4.modelEstimates/2.modelfit.do`	`FIG_flyvalid.pdf` (`4.modelEstimates/2.modelfit.do:238`)	Yes
Appendix Figure E.1: After-housing-costs income and energy poverty	`3.descriptiveResults/1.summarystats.do`	`FIG_incAHC_byinc.pdf` (`3.descriptiveResults/1.summarystats.do:393`), `FIG_energypov_bydecile.pdf` (`3.descriptiveResults/1.summarystats.do:402`)	Yes
Appendix Figure E.2: Level set of average welfare weights	`5.modelSimulations/A6.tablesfigures.do`	`FIG_lossfunctionww.pdf` (`5.modelSimulations/A6.tablesfigures.do:8`), `FIG_standardww.pdf` (`5.modelSimulations/A6.tablesfigures.do:14`), `FIG_legend_ww_ed.png` (Not found)	Yes
Appendix Figure E.3: Efficiency–targeting trade-off, under	`5.modelSimulations/A6.tablesfigures.do`	`FIG_optimal_frontier_s1.pdf` (`5.modelSimulations/A6.tablesfigures.do:20`), `FIG_optimal_frontier_s2.pdf` (`5.modelSimulations/A6.tablesfigures.do:26`), `FIG_optimal_frontier_W_legend_ed.png` (Not found)	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
more/less elastic energy demand			
Appendix Figure E.4: Social loss-minimizing policy, under more/less elastic energy demand	`5.modelSimulations/A6.tablesfigures.do`	`FIG_optimal_decomp_s.pdf` (`5.modelSimulations/A6.tablesfigures.do:31`)	Yes
Appendix Figure E.5: Counterfactual policy responses, allowing for offsetting income-based transfers	`5.modelSimulations/A6.tablesfigures.do`	`FIG_optimal_frontier_W_incadj.pdf` (`5.modelSimulations/A6.tablesfigures.do:37`), `FIG_optimal_decomp_incadj.pdf` (`5.modelSimulations/A6.tablesfigures.do:43`)	Yes
Table E.1: Public resource costs of reaching W(P0) under alternative policies	`5.modelSimulations/A6.tablesfigures.do`	`Wequiv.tex` (`5.modelSimulations/A6.tablesfigures.do:51, 54`)	Yes
Appendix Figure E.6: Variation in optimal policy with	`5.modelSimulations/A6.tablesfigures.do`	`FIG_psi_sub.pdf` (`5.modelSimulations/A6.tablesfigures.do:68`), `FIG_psi_W.pdf` (`5.modelSimulations/A6.tablesfigures.do:62`), `FIG_psi_W_nosub.pdf` (`5.modelSimulations/A6.tablesfigures.do:76`), `FIG_optimal_frontier_W_legend_ed.png` (Not found)	Yes
Appendix Figure E.7: Counterfactual policy responses, vertical equity weighting	`5.modelSimulations/A6.tablesfigures.do`	`FIG_optimal_frontier_W_incweight.pdf` (`5.modelSimulations/A6.tablesfigures.do:90`), `FIG_optimal_decomp_incweight.pdf` (`5.modelSimulations/A6.tablesfigures.do:95`)	Yes
Sample size and payment type counts	`3.descriptiveResults/1.summarystats.do`	`LOG_numusers.log` (`3.descriptiveResults/1.summarystats.do:20`)	Yes
Prepayment identification by supplier	`3.descriptiveResults/1.summarystats.do`	`LOG_likelyPP_bysupplier.log` (`3.descriptiveResults/1.summarystats.do:66`)	Yes
Energy poverty rate (2019)	`3.descriptiveResults/1.summarystats.do`	`LOG_energy_poverty.log` (`3.descriptiveResults/1.summarystats.do:380`)	Yes
LCFS prepayment share statistics	`3.descriptiveResults/1.summarystats.do`	`LOG_LCFS_ppshr.log` (`3.descriptiveResults/1.summarystats.do:723`)	Yes
Combined gas/electricity bill prevalence	`3.descriptiveResults/1.summarystats.do`	`sharecombinedbills_LCFS.txt` (`3.descriptiveResults/1.summarystats.do:743`)	Yes
Real energy price increase statistics	`3.descriptiveResults/3.describepricecap.do`	`LOG_realpriceincreases.log` (`3.descriptiveResults/3.describepricecap.do:151`)	Yes
Exposure to price shocks by income	`3.descriptiveResults/4.exposure.do`	`LOG_exposurebyincome.log` (`3.descriptiveResults/4.exposure.do:200`)	Yes
Price variation across Ofgem	`3.descriptiveResults/5.priceresponse.do`	`LOG_pvar_byregion.log` (`3.descriptiveResults/5.priceresponse.do:32`)	Yes

Figure / Table #	Program (per README)	Line Number (Actual Location for Simulated Data)	Reproduced?
regions			
MPC statistics at end of rebate period	`3.descriptiveResults/6.MPCs.do`	`LOG_MPCsendofrebates.txt` (`3.descriptiveResults/6.MPCs.do:505`)	Yes

- All tables and figures were generated successfully by the code. The outputs using synthetic data are structurally identical but numerically different from the paper, as stated by the authors. As highlighted by the authors, if synthetic dataset is used, appendix figures are created from pre-computed datasets.

9.3 In-Text Numbers

In-Text Numbers	Program	Line Number	Reproduced?
about 1.3 million household-month observations for 74,000 households	`3.descriptiveResults/1.summarystats.do`	20 (`LOG_numusers.log`)	Yes
For Boost (a prepay-only supplier), 93% of transactions are classified as likely prepayment. For Utilita… 85%.	`3.descriptiveResults/1.summarystats.do`	66 (`LOG_likelyPP_bysupplier.log`)	Yes
In 2019, we estimate that 3.9 million (14%) households are in energy poverty… more than 40% of households in the bottom income decile	`3.descriptiveResults/1.summarystats.do`	380 (`LOG_energy_poverty.log`)	Yes
14% is paid via prepayment meters… 79% of spending is paid by direct debit	`3.descriptiveResults/1.summarystats.do`	723 (`LOG_LCFS_ppshr.log`)	Yes
about 80% of electricity customers and 76% of gas customers have dual-fuel bills	`3.descriptiveResults/1.summarystats.do`	743 (`sharecombinedbills_LCFS.txt`)	Yes
real price rose by 8% in October 2021 and a further 45% in April 2022… 23% fall in July and 6% in October	`3.descriptiveResults/3.describepricecap.do`	151 (`LOG_realpriceincreases.log`)	Yes
10% of households in the bottom income decile spent more than £154 per month… average among households in the top decile (£143)… bottom income decile spent 12%… top decile 7%	`3.descriptiveResults/4.exposure.do`	200 (`LOG_exposurebyincome.log`)	Yes
average nominal cap increase was 54%, ranging from 50% in the South West to 57% in London	`3.descriptiveResults/5.priceresponse.do`	32 (`LOG_pvar_byregion.log`)	Yes

In-Text Numbers	Program	Line Number	Reproduced?
average elasticity varies from -0.25 in the bottom pre-shock spending quintile to -0.39 in the top quintile	`3.descriptiveResults/5.priceresponse.do`	818 (`LOG_elasbyeexpquint.log`)	Yes
energy budget share for prepay households (14%) over this period	`3.descriptiveResults/6.MPCs.do`	229 (`LOG_energyshr_pp_overrebates.log`)	Yes
83% of prepay households had monthly energy spending in excess of the voucher value immediately before the programme	`3.descriptiveResults/6.MPCs.do`	397 (`LOG_prop_lessthanrebate.txt`)	Yes
average MPCE out of energy-support transfers is 34%, substantially higher than the 3% MPCE from cost-of-living payments	`3.descriptiveResults/6.MPCs.do`	505 (`LOG_MPCsendofrebates.txt`)	Yes

9.4 Other Issues

- Statement about publishing data is not included, only access is described. Please include that section.
- The full reproduction took around 14 GB of space, not 5 GB as stated (because of intermediary files produced). Please adjust that information.

10 Classification

- [ ] full reproduction
- [ ] full reproduction with minor issues
- [x] partial reproduction (see above)
- [ ] not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- [ ] None.
- [ ] `Discrepancy in output` (either figures or numbers in tables or text differ)
- [ ] `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)
- [ ] `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
  - [ ] `Data preparation code missing` should be checked if the code missing seems to be data preparation code
- [ ] `Code not functional` is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- [ ] `Software not available to replicator` may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- [ ] `Insufficient time available to replicator` is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- [ ] `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- [x] `Data not available` is marked when data requires additional access steps, for instance purchase or application procedure.
- [ ] `Missing README` is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements